Weighted Heterogeneous Graph-Based Three-View Contrastive Learning for Knowledge Tracing in Personalized e-Learning Systems

Jianwen Sun, Shangheng Du, Zhi Liu, Fenghua Yu, Sannyuya Liu, and Xiaoxuan Shen

***Abstract*—Personalized e-learning systems are applications of consumer electronics in the field of education that provide individualized and adaptive services for users. Knowledge tracing (KT), as a key technology, aims to model learners' knowledge states through the interactions between learners and learning resources, and predicts their future performance. However, the problem of interaction sparsity in educational resources leads to the fact that simple representations of questions usually fail to accurately capture students' knowledge states. In this paper, inspired by the data-driven paradigm, we propose a novel knowledge tracing method named weighted heterogeneous graph-based Three-view Contrastive Learning framework for Knowledge Tracing (TCL4KT). Technically, three different view encoders in TCL4KT complement each other to obtain question embeddings with rich information. Specifically, TCL4KT considers the semantic information of higher-order, heterogeneous and the downstream task on a weighted heterogeneous graph of KT to learn high-quality representations. Besides, a meta-path-based positive sample selection strategy and joint contrastive loss are employed to gain better prediction performance. Experimental results on four datasets demonstrate the superiority of TCL4KT over baseline models, and further analysis verifies the effectiveness of our three-view contrastive learning framework.**

***Index Terms*—Heterogeneous graph, contrastive learning, knowledge tracing, personalized e-learning systems.**

I. Introduction

With the rise of digital technology, the electronic industry has produced many applications and consumer electronics in the field of education [1], [2] that have garnered significant engagement from consumers and offer them various personalized services. Personalized learning has long been a dream pursued by education, which was rated by the National Academy of Engineering as one of the 14 grand challenges to ensure human continuity [3]. Personalized e-learning system [4] is an application that combines mobile technology with online education to deliver tailored and adaptive intelligent educational services to consumers. A key aspect of personalized e-learning systems lies in their ability to accurately analyze the knowledge states of learners. Knowledge tracing (KT), as a modeling technology for learners, has become an important research content in this field [5]. KT, shown in Fig. 1, aims to estimate the current mastery states of learners through their historical learning records so as to predict the future performance, provide more targeted guidance, or make more personalized recommendations [6].

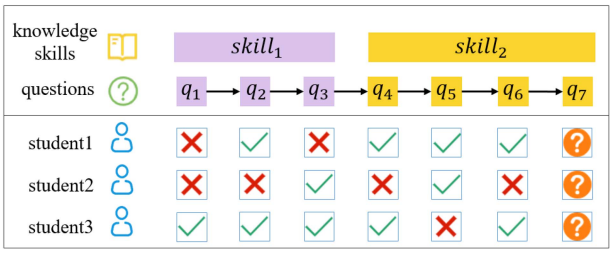

Fig. 1. A toy example of knowledge tracing task. $q_1 \rightarrow q_2 \rightarrow \cdots \rightarrow q_6 \rightarrow q_7$ is a question sequence, in which $q_1 - q_3$ examine $skill_1$, $q_4 - q_7$ examine $skill_2$, the answer interactions of three different students are recorded in the bottom black box.

The existing KT models can be categorized into the traditional KT models [6], [7] and deep learning-based KT models [8], [9], [10], [11], [12], [13], [14], [15]. In most early KT models [6], [8], [9], researchers only utilized skill information to represent each question and evaluate students' mastery of skills. These skill-level KT models often overlook the rich information contained in the questions themselves, such as difficulty and discrimination, resulting in an inability to distinguish different questions with the same skill. Therefore, in some follow-up related approaches, researchers learn the representations by incorporating various kinds of information from the questions [16], [17]. However, due to the excessive number of questions compared to the number of skills, there is a sparsity issue in the interaction between students and questions (i.e., many questions have few records of attempts), which ultimately limits the performance of the model [18].

Graph structures have proven to be highly flexible and effective in solving various complex problems, including mining potential information of data [19], [20], optimizing thread

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allocation and scheduling [21], addressing feature redundancy [22], etc. Consequently, researchers began to introduce graph structure to alleviate the problem of data sparsity by mining the corresponding relations between questions and skills (or other entities) on graphs [23], [24], [25], [26], [27]. In these graph structures, multiple questions are often associated with a single skill, while a question can belong to multiple skills simultaneously. These relations on graphs capture the similarity information of questions (e.g., the questions with the same skills may be similar), allowing the model to leverage the interaction history of similar questions and make more accurate predictions for the current question. Although these KT methods using graph structure have achieved some results, most of them only consider relations between questions and skills, and ignore the interaction between students and questions. Since the core of KT is to solve human-related problems, it is necessary to consider the information of learners in representations, so our paper constructs a heterogeneous graph containing three entities: questions, skills, and students. Different from other KT studies that introduce student entities [28], [29], due to the different types of three entities and inconsistent interaction modes, we expand the heterogeneous graph to a weighted heterogeneous graph, and employ the contrastive learning method to mine the weight information on different edges to capture higher-order semantics between questions.

Self-supervised learning technology [30] uses pretext tasks to extract supervision information from large-scale unsupervised data, and trains the network through the constructed supervision information, so as to learn valuable representations for downstream tasks. Recently, the contrastive learning (CL) method under the self-supervised framework has also been applied to the field of KT [31], [32], [33], optimizing KT tasks through joint comparative loss functions. However, these methods are only simply combined the traditional self-supervision framework and KT, and their contrastive mechanisms often ignore the characteristics of KT task itself (e.g., the heterogeneity of KT scenario, the relations between questions and skills, and the similarity of students, etc.), and even less can they consider the different relations of heterogeneous graph of KT in various perspectives.

In our paper, we propose a weighted heterogeneous graph-based three-view contrastive learning framework for knowledge tracing (TCL4KT), which exploits and integrates information of three views (meta-path, network schema and knowledge tracing) to gain question representations. Then, a special positive sample selection strategy and contrastive mechanisms are applied to maximize the mutual information between different views of the same question. Finally, we combine the encoded question embeddings and answer records to predict the future performance of students through a recurrent neural network (RNN). Compared with other baseline models, our method shows the best prediction accuracy and generalization ability on 4 common datasets. The effectiveness of our three-view contrastive learning framework is also verified through a series of analytical experiments. Our contributions are summarized as follows:

- To the best of our knowledge, this is the first time to propose a weighted heterogeneous graph-based CL framework in KT field which use three complementary view encoders integrating multiple relational information to learn more high-quality question embeddings for the prediction task.
- For GCL, we break the tradition of separating the CL framework from the downstream task during training, and propose a novel heterogeneous graph-based three-view contrastive learning framework, which uses network schema view and downstream task view to assist the meta-path view to learn node embedding, and a positive sample selection strategy and contrastive mechanism are designed to achieve better application and generalization capabilities for downstream tasks.
- We perform experiments on 4 public datasets, and the results show that the prediction performance of TC4KT is significantly improved. In addition, ablation researches verify the effectiveness of each module.

The rest of this paper is organized as follows: Section II summarizes the related work. Section III introduces TCL4KT. Section IV gives the details and results of the experiment, and Section V concludes the paper.

II. Related Works

In this part, we introduce technologies of KT including traditional KT, deep KT and the recent popular graph-based KT, and the related works of graph contrastive learning.

A. Knowledge Tracing

There are two kinds of existing knowledge tracing methods: traditional machine learning methods and deep learning methods. This paper mainly studies on the deep KT methods. In 1994, Corbett and Anderson [6] first applied knowledge tracing technology into the field of intelligent education and proposed Bayesian knowledge tracing based on hidden Markov model.

With the development of deep learning technology, deep learning-based KT models [8], [9], [11] have shown superior performance over traditional machine learning models [6]. Song et al. [34] categorize various deep KT models into DKT and its variants, memory network-based models, attention mechanism-based models, and graph-based models based on specific technologies. Piech et al. [8] employs deep learning to KT and proposes Deep Knowledge Tracing (DKT). Dynamic Key-Value Memory Networks (DKVMN) [9] utilizes a static key matrix and a dynamic value matrix to store latent knowledge skills and update the mastery of corresponding skills. Self-Attentive Knowledge Tracing (SAKT) [11] applies the self-attention mechanism [35] in KT to solve the problem of sparsity in learning interactions. Recently, the potential graph structure of KT has been explored to mine rich information, which mainly achieved by the question-skill relation to construct bipartite graphs or heterogeneous graphs [23], [24], [25], [26], [27]. In [36], a heterogeneous graph with difficulty, questions, and discrimination is constructed to capture question

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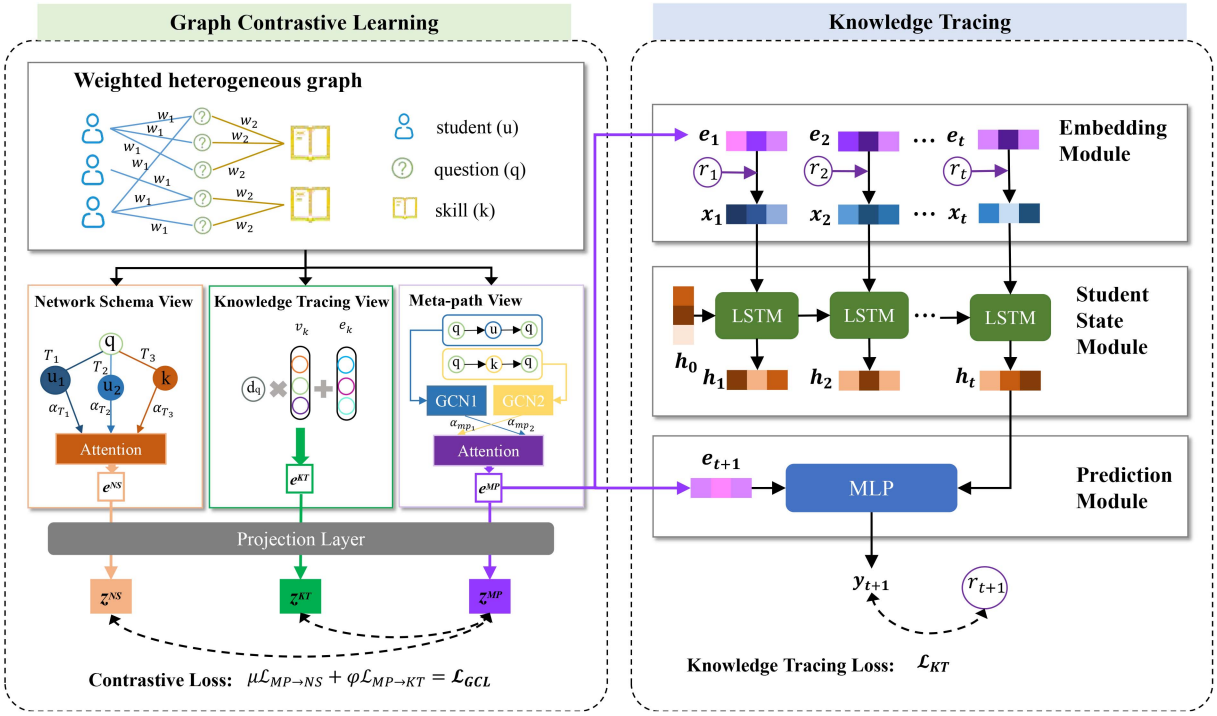

Fig. 2. Architecture of the TCL4KT model.

differences through meta-paths. Additionally, the graph convolution network (GCN) is introduced to aggregate information on the question-question or skill-skill bipartite graphs [37]. Different from them, we construct a weighted heterogeneous graph for KT, considering both skill and student information when embedding questions.

B. Contrastive Learning on Graphs

Contrastive learning is one of the main method in self-supervised learning [38], which learns the general characteristics of datasets by letting models learn the similarities or differences between data points. Graph contrastive learning (GCL) is a self-supervised learning algorithm for graph data, and it can solve the problem of traditional CL sacrificing structural information [30], [39]. For a large number of unlabeled graph data, GCL aims to train a graph encoder, such as graph neural network (GNN), to retain the characteristics of graph data. Sun et al. consider node representation and graph representation of a graph as two views, aiming to capture the consistency of local-global [40]. Wang et al. [41] proposes a co-contrastive learning method for heterogeneous GNNs, utilizing a cross-view mechanism to compound node information from different views. Inspired by it, TCL4KT proposes a three-view graph contrastive learning framework, which captures higher-order information on a heterogeneous graph of KT through three views to obtain more targeted representations of downstream tasks.

III. Proposed Method

In this section, we define some basic concepts, including weighted heterogeneous graph, meta-path and KT task, and then introduce the overall architecture of TCL4KT shown in Fig. 2, which includes a GCL part and a deep learning KT part. In the GCL part, we utilize proposed three-view CL framework to obtain question embeddings with rich semantic information. According to the existing deep learning KT models [28], [42], [43], the KT part is further divided into embedding module, student state module and prediction module.

A. Problem Formulation

1) Weighted Heterogeneous Graph: Weighted heterogeneous graph [44], [45] is defined as a graph $G = (V, E, W)$, where V and E represent sets of nodes and edges respectively, and W denotes weights on the edges, and it defines a node type mapping function $\phi{:}V \to \mathcal{A}$, an edge type mapping function $\varphi{:}E \to \mathcal{R}$ and an attribute type mapping function $\varrho{:}W \to \mathcal{U}$. $\mathcal{A}$, $\mathcal{R}$ and $\mathcal{U}$ are respectively from the corresponding network schema $S = (\mathcal{A}, \mathcal{R}, \mathcal{U})$, where $\mathcal{A} = \{A\}$, $\mathcal{R} = \{R\}$, $\mathcal{U} = \{U\}$, representing object type, edge type and attribute relation type, and $|\mathcal{A}| > 1$, $|\mathcal{R}| > 1$, $|\mathcal{U}| > 0$ in the weighted heterogeneous graph.

2) Meta-Path: A meta-path [46] is a path defined on a heterogeneous graph, and is denoted as $A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \cdots \xrightarrow{R_l} A_{l+1}$, which describes a composite relation $R = R_1 \circ R_2 \circ \cdots \circ R_l$ between object type A_1 and A_{l+1}, where $\circ$ represents the composition operator on relations.

3) Knowledge Tracing: The KT task can be formalized as: Given a student's history of learning interaction records $X_t = \{x_1, x_2, \ldots, x_t\}$, each interaction x_t is denoted as a tuple $x_t = (q_t, a_t)$, which means the student answers the question q_t at time t and gives a corresponding answer $a_t \in \{0, 1\}$ (0 means

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the answer is incorrect, 1 means correct). Similar to the current KT models, our goal is to model students' knowledge states and predict the probability $P(a_{t+1} = 1|q_{t+1}, X_t)$ of correctly answering the next question q_{t+1}.

B. Graph Contrastive Learning

1) Construction of Weighted Heterogeneous Graph: Since there are different types of nodes and interactions in the KT field, a weighted heterogeneous graph is constructed according to our training data, including three entity types: student (u), question (q), skill (k) and two weighted edges: question-student ($q-u$) and question-skill ($q-k$). In conjunction with the KT scenario, we use the correctness of students' answers as the weight (w_1) on edge $q-u$, where $w_1 \in \{0, 1\}$, and treat these two weight values as distinct relationships in subsequent calculations. The weight $w_2 \in (0, 1)$ on $q-k$ is the difficulty of the question calculated in the training data.

2) Node Feature Transformation: Considering the various node types on the weighted heterogeneous graph have different feature spaces, we first need to map them onto a common latent space using a nonlinear transformation. In particular, referring to the value of weight w_1, we divide the student node (u) into two types (u_1 and u_2), to represent students who correctly and incorrectly answer the question, respectively. Our transformation equation for a node i of type T_i is as follows:

$$p_i = \sigma\left(\mathbf{W}_{T_i} x_i + \mathbf{b}_{T_i}\right), \tag{1}$$

where p_i represents the feature after projection of node x_i, $\sigma(\cdot)$ is the nonlinear function *elu*, $\mathbf{W}_{T_i}$ denotes the type T_i specific mapping matrix, and $\mathbf{b}_{T_i}$ denotes bias.

3) Meta-Path View Encoder: The goal of this module is to learn the question embeddings with higher-order semantics from the meta-path (MP) view. Specially, to capture the potential relations between questions, we first define two weighted meta-paths $q \to u \to q$ and $q \to k \to q$ as follows:

- mp_1: $q \xrightarrow{w_1} u \xrightarrow{w_1} q$, means questions answered by the same students.
- mp_2: $q \xrightarrow{w_2} k \xrightarrow{w_2} q$, represents questions which test common skills.

Then, as a unique data enhancement technique, a biased random walk algorithm on these meta-paths is employed, with the idea being that for question nodes with the same neighbor, the closer the weight between them and the neighbor, the greater their co-occurrence probabilities are:

$$\theta_{vn} = 1 - \frac{|w_{n',v} - w_{v,n}|}{D_t}, \tag{2}$$

where θ_{vn} is the walking probability of node v to neighbor n, n and n' are the neighbors of v on the sampled meta-path, representing the previous step and the next step of v's walk, D_t is the difference between the maximum and the minimum of weight. In order to focus more on the higher-order information between questions, we mask the non-question nodes after obtaining the final sampled meta-path sequences, and use the specific meta-path-based GCN [47] to aggregate the neighbor information of questions on the ith meta-path:

$$p^{mp_i} = \left(\hat{D}^{mp_i}\right)^{-\frac{1}{2}} \hat{A}^{mp_i} \left(\hat{D}^{mp_i}\right)^{-\frac{1}{2}} p\mathbf{W}^{mp_i}, \tag{3}$$

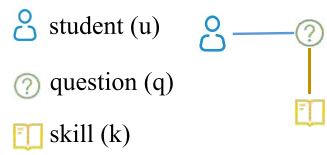

Fig. 3. Network Schema of TCL4KT.

where $p \in \{p_1, p_2, \ldots, p_n\}$ is the projective representations $\hat{D}^{mp_i}$ is the degree matrix of $\hat{A}^{mp_i}$.

Since various meta-paths represent distinct semantics on the same question, we use an attention mechanism to fuse various question representations on different meta-paths. Specifically, we first calculate the contribution from different meta-paths and convert them into weights through a *softmax* layer:

$$\begin{aligned} w_{mp_i} &= \frac{1}{|V|} \sum_{i \in V} a_{MP}^{\top} \cdot tanh\left(\mathbf{W}_{MP} \cdot p_i^{mp_i} + \mathbf{b}_{MP}\right), \\ \alpha_{mp_i} &= softmax\left(w_{mp_i}\right), \end{aligned} \tag{4}$$

where V is questions' set, $\mathbf{W}_{MP}, \mathbf{b}_{MP}$ are learnable parameters, a_{MP} is the semantic-level attention vector. Then, question embeddings on each meta-path are weighted and summed to obtain the final embeddings of this view:

$$e_i^{MP} = \sum_{n=1}^{2} \alpha_{mp_i} \cdot p_i^{mp_i}, \tag{5}$$

Furthermore, since e^{MP} considers explicit and direct relation between questions, it is finally applied to the KT module as the question representation.

4) Network Schema View Encoder: Here our goal is to learn the question embeddings considering network schema (NS), shown in Fig. 3. For common question node i, different types of neighbors (q and u) contribute differently to it, and the contributions of distinct nodes within the same neighbor type are also different.

Therefore, we first apply the graph attention mechanism (GAT) to gather messages from different nodes within a given type of neighbors:

$$\begin{aligned} \beta_{i,j}^{T_m} &= \frac{\exp\left(LeakyReLU\left(a_{T_m}^{\top} \cdot [p_i][p_j]\right)\right)}{\sum_{j \in N_i^{T_m}} \exp\left(LeakyReLU\left(a_{T_m}^{\top} \cdot [p_i][p_j]\right)\right)}, \\ h_i^{T_m} &= \sigma\left(\sum_{j \in N_i^{T_m}} \beta_{i,j}^{T_m} \cdot p_j\right), \end{aligned} \tag{6}$$

where $\beta_{i,j}^{T_m}$ is attention value from neighbor node j with type T_m to questions node i, $h_i^{T_m}$ denotes neighbor embedding with type T_m of question i, $N_i^{T_m}$ is neighbor number we random sample. The other GAT is then used to gather each type contribution. Here, $q-k$, $q-u_1$, $q-u_2$ are three kinds of different neighbor relations:

$$\begin{aligned} w_{T_m} &= \frac{1}{|V|} \sum_{i \in V} a_{NS}^{\top} \cdot tanh\left(\mathbf{W}_{NS} \cdot h_i^{T_m} + \mathbf{b}_{NS}\right), \\ \alpha_{T_m} &= softmax\left(w_{T_m}\right), \end{aligned} \tag{7}$$

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where $m \in \{1, 2, 3\}$ is type index, α_{T_m} is the importance from type T_m, $\mathbf{W}_{NS}$ and $\mathbf{b}_{NS}$ are learnable parameters.

At last, the question embedding of NS view is formed by aggregating various neighbor types:

$$e_i^{NS} = \sum_{m=1}^{S} \alpha_{T_m} \cdot h_i^{T_m}, \tag{8}$$

where S is the total number of neighbors' types.

5) Knowledge Tracing View Encoder: In this view, we hope to combine the latest progress of the downstream task (i.e., KT) to help encode question embeddings. Inspired by AKT [13], we consider the difficulty of questions and the unique state of their corresponding skills for better representations:

$$e_i^{KT} = e_{k_i} + d_{q_i} \cdot v_{k_i}, \tag{9}$$

where $e_{k_i} \in 1 \times d$ denotes the ith skill's state that summarizes the variation in questions covering it, d_{q_i} is the difficulty parameter, $v_{k_i} \in 1 \times d$ is the ith skill embedding. Finally, we get e^{KT} as the embedding of KT view.

6) Contrastive Loss: In order to calculate the contrastive loss of our three-view framework, we apply a projection layer to map question representations of different views to the same latent space, for each view:

$$Z_i^{\Omega} = \mathbf{W}_2 \cdot \sigma\left(\mathbf{W}_1 \cdot e_i^{\Omega} + \mathbf{b}_1\right) + \mathbf{b}_2, \tag{10}$$

where we use $\Omega = \{MP, KT, NS\}$ to denote different view types. $\mathbf{W}_1, \mathbf{W}_2, \mathbf{b}_1, \mathbf{b}_2$ are common learnable parameters, and *elu* is used as the nonlinear activation function σ. Then, we apply a positive sample selection strategy based on meta-paths of MP view: Given a target node i, we count the number of occurrences of questions adjacent to i on all meta-paths. After descending sorting, nodes with occurrence times $\geq K$ (K is the manually set threshold) will be selected as the positive samples' set P_i of target node i, and other remaining nodes are considered as negative samples' set N_i. According to [48], we regard MP view as the core view, and calculate the contrastive loss between it and NS view, KT view respectively. In the contrastive part of MP-NS, we select target embeddings from MP view and positive and negative sample embeddings from NS view to calculate the loss in MP view of MP-NS part:

$$L_i^{MP} = -\log \frac{\sum_{j \in P_i} exp\left(sim\left(Z_i^{MP}, Z_j^{NS}\right)/\tau\right)}{\sum_{k \in \{P_i \cup N_i\}} exp\left(sim\left(Z_i^{MP}, Z_k^{NS}\right)/\tau\right)}, \tag{11}$$

where P_i and N_i are the positive and negative samples selected above, and *sim* is the cosine similarity calculation function, τ is a temperature parameter.

Similar to the procedure of calculating L_i^{MP}, the target embedding is from NS view (Z_i^{NS}), while the embeddings of positive and negative samples are from MP view (Z_k^{MP}). Combined with the cross calculation of the two views, we get the final loss of MP-NS ($L_{MP \to NS}$):

$$L_{MP \to NS} = \frac{1}{|V|} \sum_{i \in V} \left[\lambda_1 \cdot L_i^{NS} + (1 - \lambda_1) \cdot L_i^{MP}\right], \tag{12}$$

Similarly, the contrastive loss of MP-KT ($L_{MP \to KT}$) can also be obtained according to the same calculation as $L_{MP \to NS}$:

$$L_{MP \to KT} = \frac{1}{|V|} \sum_{i \in V} \left[\lambda_2 \cdot L_i^{KT} + (1 - \lambda_2) \cdot L_i^{MP}\right], \tag{13}$$

where $L_{MP \to NS}$, $L_{MP \to KT}$ are the final contrastive loss of MP-NS and MP-KT respectively, λ_1, λ_2 are user-defined balance coefficients.

C. Knowledge Tracing

1) Embedding Module: This module aims to model the interaction of students' history exercises for KT model. At each historical time t, the question embedding from MP view ($e_t = e^{MP}$) and a d-dimensional zero vector are concatenated to obtain the final interaction vector $x_t \in R^{2d}$:

$$x_t = \begin{cases} [e_t \oplus 0], & r_t = 1 \\ [0 \oplus e_t], & r_t = 0. \end{cases} \tag{14}$$

2) Student State Module: Like most deep learning-based KT models [8], [23], [27], we model the student's knowledge state h_t using a sequence model (LSTM) based on the current input x_t and the previous state h_{t-1}, and the calculation formula is as follows:

$$h_t = LSTM(x_t, h_{t-1}, \theta), \tag{15}$$

where θ means the learnable parameters in LSTM model.

3) Predict Module: Students' knowledge state vector h_t and the embedding of next question e^{t+1} are used to predict the probability of students answering the next question correctly y_{t+1}. Here, we estimate the interaction function between h_t and e^{t+1} using a multi-layer perceptron (MLP):

$$\begin{aligned} \gamma_{t+1} &= \sigma\left(\mathbf{W}_1 \cdot \left[h_t \oplus e_{t+1}\right] + \mathbf{b}_1\right), \\ y_{t+1} &= sigmoid(\mathbf{W}_2 \cdot \gamma_{t+1} + \mathbf{b}_2), \end{aligned} \tag{16}$$

where γ_{t+1} is a comprehensive representation of the next question and the current student's knowledge state, $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{b}_1$, $\mathbf{b}_2$ are learnable parameters.

Finally, the loss function of the KT part is defined as the binary cross entropy loss between the prediction probability and the true label of the student' answers from step 1 to T. At t-th step, let r_t be the actual score and y_t be the predicted score through TCL4KT on the t-th question q_t:

$$L_{KT} = -\sum_{t=1}^{T} (r_t \log y_t + (1 - r_t) \log(1 - y_t)). \tag{17}$$

D. Model Training

The total objective function of TCL4KT is a linear combination of GCL loss ($L_{MP \to NS}$ and $L_{MP \to KT}$) and KT loss:

$$L = \mu L_{MP \to NS} + \varphi L_{MP \to KT} + L_{KT}, \tag{18}$$

where μ, φ are the hyperparameters that control the influence of the self-supervised learning signal. In the training phase, Adam gradient descent algorithm is used to minimize (18).

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TABLE I
SUMMARY STATISTICS OF PROCESSED DATASETS

	ASSIST09	ASSIST17	EdNet	Statics2011
# student (u)	4,160	1,706	5,002	331
# question (q)	15,680	1,150	11,775	633
# skill (k)	167	86	1,837	106
# $q-u$	207,659	153,324	1,156,254	91,133
# $q-k$	189,370	1,150	26,345	733

IV. EXPERIMENTAL RESULTS

In this section, we evaluate the performance of TCL4KT model and verify the effectiveness of three-view contrastive learning. We first introduce the datasets and experimental setting in Sections IV-A and IV-B, and our experimental results are shown in Sections IV-C–IV-F.

A. Datasets

Four real datasets are used in our experiments: ASSIST09,[1] ASSIST17,[2] EdNet[3] and Statics2011.[4] ASSIST09 and ASSIST17 are the ASSIST series datasets from the ASSISTments online tutoring platform, of which ASSIST09 that collected from the 2009-2010 academic year, is the most commonly used benchmark dataset in the KT field. EdNet is a large-scale hierarchical dataset in education, which collected by the online tutoring platform Santa from 2017 to 2019. Statics2011 contains student responses to exercises from the Fall 2011 Engineering Statics course at Carnegie Mellon University (CMU). Referring to the existing research work [10], [13], [23], [49], we preprocess the datasets as follows: Scaffold questions and answer records without skills are removed [13]. For skill-level models, we combine multiple skills into a new joint skill [49]. For large-scale datasets like EdNet, we randomly sample data from 5000 students [23]. For all datasets, we randomly split the data into 80% training data and 20% test data. When loading data, we remove the answer sequences with a length shorter than 3, because too short sequences are meaningless, and those sequences with a length longer than 200 are divided into several shorter ones to speed up the calculation efficiency [13]. Table I shows the statistics of the processed datasets.

B. Experimental Settings

1) Metrics: Since our downstream application, knowledge tracing, is a binary classification problem, same as the previous work [12], [32], [37], Area Under the Curve (AUC) and Accuracy (ACC) are used to evaluate the predictive performance of models.

2) Baselines: In order to evaluate the performance of TCL4KT, we compare the following models as our baselines:

- **BKT** [6] is the first probability-based KT model that models each skill individually based on hidden markov model.
- **DKT** [8] is the original deep learning-based KT model that applies recurrent neural network (RNN).
- **SAKT** [11] is the first Self-Attention-based Knowledge Tracing model, which uses transformer to solve the problem of long-term dependence.
- **GIKT** [23] is a Graph-based Interaction model for Knowledge Tracing using graph convolution neural network (GCN) to combine question-skill correlations.
- **GASKT** [25] is a Graph-based Attentive Knowledge-Search model for Knowledge Tracing, which considers both heterogeneity and forgetting factors.
- **CoKT** [28] construct two bipartite graphs to capture the relationship between questions and students (skills), and gain Collaborative embeddings for Knowledge Tracing in the pre-training module.
- **AKT** [13] is a Context-Aware Attentive Knowledge Tracing model, which couples the attention-based neural network model with the interpretable model components.
- **PEBG+DKT** [24] improves knowledge tracing via Pre-training question Embeddings on Bipartite Graphs, which utilizes question embeddings pre-trained by PEBG and make DKT achieve question-level prediction.
- **CL4KT** [31] is the model to alleviate the problem of sparse interactions between students and questions, which introduces a Contrastive Learning framework for Knowledge Tracing for the first time.
- **CDKT** [33] introduces Contrastive learning framework into DKT model for gaining a high-quality representation of students.

3) Implementation Details: We set the hyper-parameters of TCL4KT as follows: In MP view, the number of walks per node is 10, the walk length is 80, the window size is 3. For neighbor sampling in (6) in NS view, the number of skills is 1, students who answer the question correctly and incorrectly are selected as 5 and 1 respectively. In contrastive loss, threshold $K \in \{1, 2\}$, τ is set to 0.8. The selection of μ and φ is obtained by grid search, as detailed in Section IV-F. For model training, the dimension of embeddings is 128, batch size is 80, learning rate is 0.008. We add L2 norm on the weights and set weight decay as 1e-5. Dropout in (16) is set to 0.3. For other models, their hyper-parameters are either set as the original optimal setting in their papers or optimized by grid search. All models are implemented with Pytorch using Python on a Windows server that equips with 64GB RAM, sixteen Intel (R) i9-9900K CPUs and two GeForce RTX 2080 Ti GPUs.

C. Performance Comparison

We repeat each experiment for 5 times, and record the average AUC and ACC of all models in Table II. From the experimental results, we can draw the following conclusions: First, all baseline models based on deep learning outperform conventional machine learning-based BKT in terms of performance, demonstrating the effectiveness of deep learning method. Next, compared with the origin DKT that only considers the question (or skill) information, those models that use graph method to mine the question-skill relations, such as GIKT, GASKT, CoKT and PEBG, have achieved good

[1]https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data
[2]https://sites.google.com/view/assistmentsdatamining/dataset
[3]https://github.com/riiid/ednet
[4]https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT MODELS REGARDING AUC AND ACC

Model	ASSIST09		ASSIST17		EdNet		Statics2011	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
BKT	0.7180	0.6922	0.6519	0.6290	0.6737	0.7018	0.7444	0.7973
DKT	0.7676	0.7286	0.7186	0.6674	0.7015	0.7124	0.8403	0.8270
SAKT	0.7577	0.7256	0.7077	0.6651	0.6895	0.7124	0.8252	0.8104
GIKT	0.7859	0.7371	0.7723	0.6989	0.7640	0.7366	0.8907	0.8503
GASKT	0.7781	0.7362	0.7636	0.6939	0.7647	0.7374	0.8786	0.8449
CoKT	0.7859	0.7393	0.7637	0.6972	0.7654	0.7387	0.8746	0.8436
AKT	0.7888	0.7427	0.7620	0.6937	0.7647	0.7385	0.8718	0.8376
PEBG+DKT	0.7738	0.7329	0.7619	0.6949	0.7571	0.7366	0.8655	0.8368
CL4KT	0.7626	0.7275	0.7139	0.6615	0.6965	0.7118	0.8251	0.8170
CDKT	0.7733	0.7297	0.7709	0.7019	0.7645	0.7386	0.8872	0.8510
TCL4KT	**0.7927**	**0.7429**	**0.7728**	**0.7043**	**0.7663**	**0.7388**	**0.8933**	**0.8545**

TABLE III
AUC COMPARISON OF THREE VIEW ENCODERS

View	ASSIST09	ASSIST17	EdNet	Statics2011
NS View	0.7630	0.7487	0.7344	0.8650
KT View	0.7524	0.7634	0.7597	0.8870
MP View	**0.7861**	**0.7722**	**0.7652**	**0.8916**

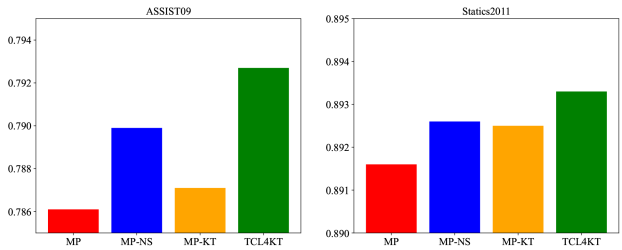

Fig. 4. Performance comparison on different view combinations.

promotion. Then, when comparing the performance of SAKT and AKT, it seems that the better effect of AKT in attention mechanism can also be attributed to the capture of the question-skill relations. Finally, our TCL4KT model that uses a three-view contrastive learning framework considering various information between skills, students and questions on a weighted heterogeneous graph to encode question embeddings with rich semantics, achieves the best performance on all four datasets, which proves the effectiveness and superiority of our method. In addition, among the contrastive learning based KT models, CDKT has achieved significant improvement by using a simple CL mechanism on the DKT, reflecting the effectiveness of introducing CL method. The average performance of CL4KT inputting skill information is far weaker than TCL4KT and CDKT that use question information as embeddings on AUC and ACC, indicating that using only skills as input is insufficient, which ignores valuable and rich question information.

D. Ablation Study

In this section, we report the results of ablation studies. In order to verify the role of the three-view contrastive learning framework, the view encoders will be further explored. On the one hand, in Table III, we compare the AUC of NS view, KT view and MP view as the only single view encoder to judge which one is more suitable for our KT task. We can observe that it is obvious that among the three view encoders, MP view performs best on all datasets. Specially, TCL4KT with single MP view encoder respectively lead encoder with NS view and KT view by 1.04% and 1.32% on AUC on average, which explicitly aggregates neighbors and considers higher-order semantic information via weighted meta-paths, and also demonstrates the superiority of using the MP view as the question representations of the KT downstream task. In addition, KT view and NS view have inconsistent results on various datasets, indicating that the semantics and emphasis of different views are distinct, and the distribution of datasets can affect the representation ability of the view encoders.

On the other hand, we replace the three-view contrastive framework in our TCL4KT model with single view embeddings of MP view and dual view contrastive embeddings for MP-NS and MP-KT. Fig. 4 shows the research results on ASSIST09 and Statics2011. We can find that: Firstly, the effect of using single MP view encoder as the model embeddings method is lower than that of using cross-view contrastive learning methods (MP-NS and MP-KT), and TCL4KT achieves the best performance by using three-view CL framework, which shows that the more views considered, the better the performance of question embeddings. Secondly, for different datasets, the increase of results caused by adding new views is inconsistent, which shows that the latent relations of views and datasets are closely related. Finally, the effect of MP-KT is not as good as that of MP-NS. This may be because when combining with MP view, the information about questions and skills specially considered in the KT view is potentially already learned by the model in the NS view, which learns network schema information of questions.

E. Visualization Analysis

We employ t-SNE to visualize the question embeddings learned by CL4KT in a 2D space. The top 15 skills, covering the most questions, are selected and shown in the legend of Fig. 5 on the ASSIST09 dataset. Fig. 5 shows a visualization of question embeddings related these 15 skills, where questions involving the same skill are indicated with the same

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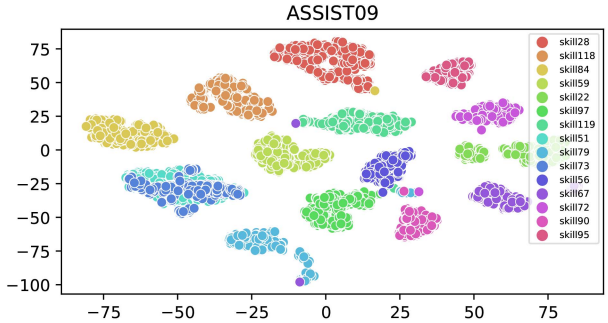

Fig. 5. Visualization of question embeddings.

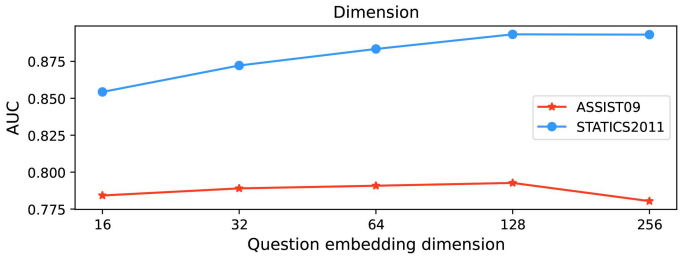

Fig. 6. Performance comparison of different question embedding dimensions.

color, revealing tightly clustered embeddings. Besides, questions with different skills are well separated, which proves the close relationship between questions and also reflects the effectiveness of our contrastive learning framework to extract information.

F. Hyperparameter Exploration

Due to the inseparable relationship between parameter choice and model accuracy, we further explore the embedding dimension of question and hyerparameters of balancing contrastive loss in (18) to obtain the best accuracy of the model.

1) Impact of Question Embeddings Dimension: The question embeddings dimension = {16,32,64,128,256} are tried in the experiment to explore the best performance of the model. According to the Fig. 6, we can clearly see that the change trends of model performance with various embedding dimensions on the ASSISIT09 and Statics2011 datasets are consistent. As the embedding dimension increases, performance of TCL4KT improves first and then degrades, which is because the model is under-fitting while the embedding dimension is small, resulting in low model accuracy. When the embedding dimension exceeds the threshold (128), the model accuracy begins to decline due to over-fitting of model. Therefore, in this paper, we set the embedding dimension = 128 for TCL4KT to achieve the best performance.

2) Impact of Graph Contrastive Loss: The regularized weights μ and φ controlling the contrastive loss of MP-NS and MP-KT respectively in (18), are the important hyperparameters that affect TCL4KT loss. To analyze the impact of GCL loss, we examine the influence of $L_{MP\to NS}$ and $L_{MP\to KT}$ by changing μ and φ. We try $\mu = \{0.01, 0.05, 0.1, 0.5\}$, $\varphi = \{0, 0.0005, 0.005, 0.05, 0.5, 5\}$ to explore the hyperparameter combination of TCL4KT to achieve the best performance.

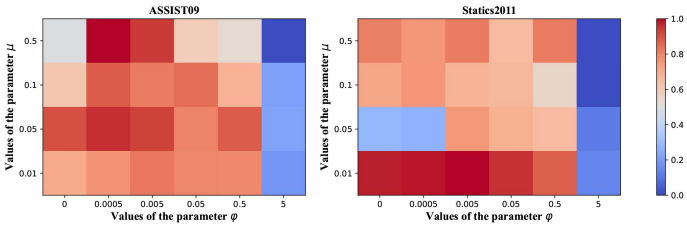

Fig. 7. Performance comparison with respect to μ and φ.

As shown in Fig. 7, we can find that: For most values of μ, the trend of AUC value increases with the increase of φ, then fluctuates to a high point gradually, which indicates the effectiveness of MP and KT views construction, and finally decreases rapidly when reaching a large threshold, because the loss of GCL should not overwhelms L^{KT} of our downstream task. For Statics2011, a dataset with a small number of questions, the better results are obtained from the smaller μ and φ, which may be to avoid the risk of over fitting the model. Excessive μ and φ will greatly increase the impact of GCL loss, which is not conducive to the generalization of the model. For ASSIST09 with many questions, since the cross contrastive loss of MP-NS can promote the acquisition of high-quality embeddings, a larger μ and a smaller φ mean a higher AUC value, and better generalization can be achieved by fine-tuning the contrastive loss $L_{MP\to KT}$. In a word, the generalization ability of the model and prediction performance are improved through the regulation of the three-view contrastive loss.

V. Conclusion

In this paper, we propose a weighted heterogeneous graph-based three-view contrastive learning framework for knowledge tracing. For graph contrastive learning, we fully capture the information on the weighted heterogeneous graph through three mutually coordinated and supervised view encoders, which break the tradition of separating the CL framework from the downstream tasks in the training process. For the downstream task KT, by integrating semantic information from three views of constructed weighted heterogeneous graph, we enhance question representation and achieve more accurate predictions of learners' future performance, which will help the e-learning systems provide more personalized services to consumers, such as resource recommendation and individual learning plan making. Experiments on four datasets show that TCL4KT outperforms existing KT models, and ablation study and visualization experiments further verify the effectiveness of proposed three-view contrastive learning framework.

However, there still exist some limitations in the proposed framework that need further exploration. For instance, in construction of weighted heterogeneous graph, it may cause subjective deviation to only regard only questions' difficulty through data statistics as the weight and consider that entities with similar weights are related. Additionally, in student state module, it may not be enough to simply use DKT model applying LSTM to analysis learns' states. Therefore, in future work, the weighted heterogeneous graph representation could

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be further expand and improve. Specifically, we can introduce additional node attribute or employ betweenness centrality to quantify the global significance of different vertices [50]. Furthermore, clustering the entities on graphs [51] as alternatives to data statistical approaches may enhance the precision of edge weights. Besides, with the development of KT models, we could incorporate more novel structures into the student state module to better estimate learners' states, leading to improved predictions and interpretability.

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